Formula Mass and Percent Composition

Formula mass, mass percent and mass of an element in a sample

Name _____ Class _____ Date _____

DATA

H = 1.008 u C = 12.01 u N = 14.01 u O = 16.00 u Na = 22.99 u Mg = 24.31 u

S = 32.06 u Ca = 40.08 u Cu = 63.55 u

Show your working. Give percentages to two decimal places.

A. Understanding

1. What does the percent composition of a compound tell you?

2. In methane, CH_4, four of the five atoms are hydrogen, yet hydrogen is only about 25% of the mass. Explain why.

B. Formula mass

3. Calculate the formula mass of (a) H_2SO_4, (b) $Mg(OH)_2$, (c) $(NH_4)_2SO_4$.

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C. Percent composition

4. Calculate the percent composition by mass of sodium hydroxide, $NaOH$.

5. Calculate the mass percent of nitrogen in ammonium nitrate, NH_4NO_3.

6. Calculate the percent composition by mass of ethanol, C_2H_5OH. Check that your percentages add up to 100%.

D. Mass of an element in a sample

7. What mass of copper is contained in 25.0 g of copper(II) sulfate, $CuSO_4$?

8. What mass of oxygen is contained in 50.0 g of calcium carbonate, $CaCO_3$?

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E. Challenge

9. Which fertiliser contains the higher mass percent of nitrogen: ammonia, NH_3, or ammonium sulfate, $(NH_4)_2SO_4$? Show your calculations.

10. Copper(II) sulfate crystals are a hydrate, $CuSO_4 \cdot 5\,H_2O$. What percentage of the hydrate's mass is water?

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